DESIGN AND ANALYSIS OF ALGORITHM

LAB EXPERIMENT -7

NAME:

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ROLL NO:

CH.SC.U4CSE24224

1. JOB SCHEDULLING CODE:

```
  GNU nano 7.2
#include <stdio.h>

struct Job{
    char id;
    int deadline;
    int profit;
};

void swap(struct Job *a, struct Job *b){
    struct Job temp = *a;
    *a = *b;
    *b = temp;
}

void sortJobs(struct Job arr[], int n){
    for(int i=0;i<n-1;i++){
        for(int j=0;j<n-i-1;j++){
            if(arr[j].profit < arr[j+1].profit)
                swap(&arr[j],&arr[j+1]);
        }
    }
}

void jobScheduling(struct Job arr[], int n){
    sortJobs(arr,n);

    int result[n];
    int slot[n];

    for(int i=0;i<n;i++)
        slot[i]=0;

    for(int i=0;i<n;i++){
        for(int j=arr[i].deadline-1;j>=0;j--){
            if(slot[j]==0){
                result[j]=i;
                slot[j]=1;
                break;
            }
        }
    }

    printf("Job sequence: ");
    for(int i=0;i<n;i++)
        if(slot[i])
            printf("%c ",arr[result[i]].id);
```

```c
}

int main(){

    printf("Name : Kolathur Surya\n");
    printf("Roll No : CH.SC.U4CSE24224\n\n");

    struct Job arr[]={
        {'A',2,100},
        {'B',1,19},
        {'C',2,27},
        {'D',1,25},
        {'E',3,15}
    };

    int n = sizeof(arr)/sizeof(arr[0]);

    jobScheduling(arr,n);

    return 0;
}
```

OUTPUT:

```
surya@DESKTOP-FJ2EKA4:~$ nano job.c
surya@DESKTOP-FJ2EKA4:~$ gcc job.c -o job
surya@DESKTOP-FJ2EKA4:~$ ./job
Name : Kolathur Surya
Roll No : CH.SC.U4CSE24224

Job sequence: C A E surya@DESKTOP-FJ2EKA4:~$
```

INTERPRETATION:

Job Scheduling with deadlines is a greedy algorithm used to schedule jobs in a way that maximizes total profit while completing each job before its deadline.
It works by sorting jobs in decreasing order of profit and assigning them to the latest available time slot before their deadline.